Arjun Konidala

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.73
Narayana Junior College	Class XII (IPE)	2023	97.60%
Narayana EM High School	Class X (BSEAP)	2021	100.00%

SKILLS

Programming Languages: C++, C, Python

Database Management: SQL

Tools: Linux, Git, Bash Scripting, Latex, MySQL, Make, GDB, awk

Competitive Programming: Codeforces

WORK EXPERIENCE

Advance Developer | *Samsung Research Institute Bengaluru -Beyond 5G (B5G) Team*

- Engineered a **Transformer-based architecture** in PyTorch to predict scene-aware wireless channel parameters using multi-angle camera feeds and contextual spatial location data to reduce latency by 50% relative **Sionna RT.**
- **Integrated Tx/Rx spatial embeddings** into visual features using 3D perspective projection, accurately mapping physical transmitter and receiver coordinates directly onto multi-view image frames.

PROJECTS

CareFlow: AI Care Coordination Assistant

- Engineered **LangGraph-based multi-agent workflows** using state-machine orchestration for specialized agents.
- Built a **production RAG pipeline** grounded in clinical guidelines with guardrails for unsafe diagnostic requests.
- Integrated external tools via **MCP**, enabling secure real-time updates to databases and communication platforms.
- Implemented **end-to-end tracing and observability** for model accuracy, task completion, latency, and reliability.

Multi-Client Quiz System with LLM Integration | (Course Work Prof. Kotaro / Prof. Praveen)

- Developed a **concurrent TCP quiz server in C++** integrating an LLM for dynamic question generation and answer evaluation.
- Implemented a **custom application-layer protocol** over TCP with message framing and **keep-alive mechanisms**.
- Added **TTL-based caching** to serve repeated quiz requests efficiently while limiting unnecessary LLM calls.
- Profiled **network throughput and request latency** across different client loads using **Wireshark**.

Memory Management in GEMOS(Course Work (CS3523), Prof. Saurabh Kumar)

- Engineered mmap, munmap, and mprotect system calls to manage user-space virtual memory mappings.
- Developed pagefault handler which enables lazy allocation of physical memory for optimal usage of RAM memory and mimicking real OS behavior.

Synchronization in Multithreaded Programs (Course Work (CS3523), Prof. Sathya Peri)

- Implemented busy-wait synchronization using Test-and-Set (TAS) and Compare-and-Swap (CAS) atomic instructions.
- Benchmarked synchronization performance by measuring critical-section latency across varying thread loads.

Assembly Instructions Simulator (Course work /Prof. Rajesh Kedia)

- Designed and implemented a Simulator using C++ that simulates the assembly instructions sequentially with features like break points, stack display, registers view and memory display are included in the simulation.
- Customizable cache memory is simulated (associativity , block size , capacity and replacement and write policies).Tracked hit/miss statistics to optimize cache dimensions and replacement policies for workload-specific performance.
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RELEVANT COURSES

CSE Core Courses: Foundations of Machine Learning, Deep Learning, Computer Networks, Software Development Fundamentals, Data Structures and Algorithms, Computer Architecture, Operating Systems, DBMS, Discrete Mathematics, Algorithmic Techniques for Large Graphs, Theory of computation

Mathematics & EE Core Courses: Probability & Random Variables, Introduction to Statistics, Calculus, Linear Algebra

SCHOLASTIC ACHIEVEMENTS

- Secured a position in the top 50 regionally and qualified IOQM 2021 conducted by MTA, India.

POSITIONS OF RESPONSIBILITY

- **Placement Coordinator** at OCS, IITH, pitching the institute to prospective companies and coordinating events.
- TA for the course DSA, Theory of computation and AI. Guided students during labs and evaluated answer scripts.
- Web coordinator, Alumni Cell IIT Hyderabad 2025 - 2026
- BlockSpace Core, Blockchain Technology Club IIT Hyderabad

EXTRACURRICULAR

- NSO member of Hockey
- Badminton and Gym enthusiast